



can you list out the fields for property cdm

It sounds like you might be asking about the data fields you need to record to comply with the Construction (Design and Management) Regulations 2015 (CDM 2015) for a property or project. ^[1]

Because “property cdm” is a bit ambiguous, I’ll list the typical fields people include when they set up a CDM record or form for a property or project (for example in a property management or construction management system in the UK).

Core project fields

- Project / property name ^[1]
- Property address (including postcode) ^[1]
- Client name (dutyholder under CDM) ^[1]
- Client type (commercial or domestic) ^{[2] [1]}
- Project description / scope of works ^[2]
- Project reference / job number ^[2]
- Project start date and anticipated completion date ^[2]

CDM dutyholder fields

For CDM 2015, you usually capture who is filling each role: ^{[2] [1]}

- Principal Designer name
- Principal Designer organisation
- Principal Designer contact details (phone, email)
- Principal Contractor name
- Principal Contractor organisation
- Principal Contractor contact details
- Any other Designers (names / organisations)
- Other Contractors (names / organisations)

HSE notification (F10) fields

If the project is notifiable, you typically store:^[1] ^[2]

- Whether the project is notifiable (yes / no)
- Date of F10 notification
- HSE notification reference number
- Person who submitted F10
- Copy / link to F10 form

Health and safety documentation fields

To show that CDM duties are being met, systems often record:^[2] ^[1]

- Pre-construction information (location / file link, date issued)
- Construction phase plan (status, date approved, file link)
- Health and safety file (required yes/no, status, file link)
- Risk assessments and method statements (RAMS) references / links
- Records of design risk reviews / design meetings
- Site inductions completed (yes/no, date started)
- Welfare facilities arrangements (description / confirmation)

Compliance and status tracking fields

To manage ongoing compliance across a property or portfolio:^[3] ^[1] ^[2]

- CDM applicability (does CDM 2015 apply to this work?)
- Number of contractors involved (single / multiple)
- Principal Contractor formally appointed (yes/no, date)
- Principal Designer formally appointed (yes/no, date)
- Key CDM duties checked (tick boxes, e.g. “sufficient time and resources allowed”, “appointments in writing”, “competence checked”)
- Last CDM review date
- CDM responsible person within your organisation
- Notes / actions / outstanding issues

If by “property cdm” you meant something more specific (for example a particular software system, database schema, or a template from your organisation), what system or context are you working in?



these .py define the current CDM. can you list out the available fields

Your current Property CDM defines a nested schema (PROPERTY_SCHEMA) and a flat snake_case mapping for downstream use. I'll list both, since they are both “available fields” in practice.^[16] ^[17]

1. Nested CDM fields (PROPERTY_SCHEMA)

These are the canonical fields under PROPERTY_SCHEMA grouped by section.^[17]

PropertyHeader.Header

- UPRN
- PropertyID
- CatchmentID
- propertyType
- propertyStatus

PropertyHeader.Valuation

- PropertyValue
- ValuationDate
- ValuationMethod

PropertyHeader.PropertyAttributes

- OccupancyType
- PropertyAreaSqm
- PropertyResi
- NumberOfStoreys
- ConstructionYear
- PropertyPeriod
- CouncilTaxBand
- NumberBedrooms
- NumberBathrooms
- PropertyCondition

PropertyHeader.Construction

- ConstructionType
- FoundationType
- FloorLevelMeters
- BasementPresent

PropertyHeader.Location

- BuildingNumber
- StreetName
- TownCity
- County
- Postcode
- LocalAuthority
- Country
- Region
- UrbanRuralClassification
- LatitudeDegrees
- LongitudeDegrees

PropertyHeader.RiskAssessment

- EAFloodZone
- OverallFloodRisk
- FloodRiskType
- GroundLevelMeters
- RiverDistanceMeters

ProtectionMeasures.RiskAssessment

- InsurancePremium
- RiskRating

ProtectionMeasures.ResilienceMeasures

- FloodGates
- FloodBarriers
- SumpPump
- FloodWarningSystem

TransactionHistory.Purchase

- PurchaseDate
- PurchasePriceGbp

TransactionHistory.Rental

- MonthlyRentGbp
- RentalYield

The validator also treats these as required (or recommended) at minimum:^[18]

- PropertyHeader.Header.PropertyID
- PropertyHeader.Header.CatchmentID
- PropertyHeader.Location.LatitudeDegrees
- PropertyHeader.Location.LongitudeDegrees

2. Flat mapped fields (create_mapping output)

`create_mapping` flattens a nested CDM record into a snake_case dict and strips None values. These are the output keys:^[16]

Header / IDs

- property_id
- catchment_id
- uprn
- property_type
- property_status

Valuation

- value
- valuation_date
- valuation_method

Property attributes

- occupancy_type
- property_area_sqm
- property_resi
- number_storeys
- construction_year
- property_period

- council_tax_band
- number_bedrooms
- number_bathrooms
- property_condition

Construction

- construction_type
- foundation_type
- floor_level_metres
- basement_present

Location

- building_number
- street_name
- town_city
- county
- postcode
- local_authority
- country
- region
- urban_rural
- latitude
- longitude

Risk / flood

- flood_zone
- overall_flood_risk
- flood_risk_type
- ground_level_meters
- elevation (alias of ground_level_meters)
- river_distance

Protection / resilience

- insurance_premium
- risk_rating
- flood_gates
- flood_barriers

- sump_pump
- flood_warning_system

Transactions

- purchase_date
- purchase_price
- monthly_rent
- rental_yield

3. Quick table: nested → flat

Concept	Nested CDM field	Flat key	
Property ID	PropertyHeader.Header.PropertyID	property_id	
Catchment ID	PropertyHeader.Header.CatchmentID	catchment_id	
Latitude	PropertyHeader.Location.LatitudeDegrees	latitude	
Longitude	PropertyHeader.Location.LongitudeDegrees	longitude	
Ground level	PropertyHeader.RiskAssessment.GroundLevelMeters	ground_level_meters	
EA flood zone	PropertyHeader.RiskAssessment.EAFloodZone	flood_zone	
Insurance premium	ProtectionMeasures.RiskAssessment.InsurancePremium	insurance_premium	
Purchase price (GBP)	TransactionHistory.Purchase.PurchasePriceGbp	purchase_price	
Monthly rent (GBP)	TransactionHistory.Rental.MonthlyRentGbp	monthly_rent	[16] [17]

Do you mainly care about the nested CDM schema (for validation / UI), or the flat mapping keys (for the model / DB), so I can format this into something like a JSON schema or SQL DDL for you?

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from the BRI information, what fields are required

From the code you shared, the only fields that are explicitly marked as required in the current Property CDM/BRI validation are these dotted paths:[\[20\]](#)

- PropertyHeader.Header.PropertyID
- PropertyHeader.Header.CatchmentID (labelled “recommended” in the error message, but still included in `get_required_fields()`)
- PropertyHeader.Location.LatitudeDegrees
- PropertyHeader.Location.LongitudeDegrees

These four are what `get_required_fields()` returns and what `validate()` enforces with “Missing required field ...” (or “recommended” in the message for CatchmentID) when absent.[\[20\]](#)

To confirm: when you say “from the BRI information”, are you treating those four as the mandatory minimum, or do you expect more BRI-specific fields to be required that just aren’t enforced in this validator yet?

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OK, so what are BRI specific fields

Within the code you shared, there is nothing explicitly named “BRI” or “Building Resilience Index”, so there are no fields literally tagged as BRI-only. However, the schema clearly includes a subset of fields that are resilience/risk focused and are the ones you would typically use for any BRI-style or resilience index scoring.^[33] ^[34] ^[35]

Risk and resilience (BRI-relevant) fields

These are the fields in your Property CDM that are clearly about risk, exposure, and resilience – ie. the natural candidates for BRI-specific use.^[35]

- PropertyHeader.RiskAssessment.EAFloodZone
- PropertyHeader.RiskAssessment.OverallFloodRisk
- PropertyHeader.RiskAssessment.FloodRiskType
- PropertyHeader.RiskAssessment.GroundLevelMeters
- PropertyHeader.RiskAssessment.RiverDistanceMeters
- ProtectionMeasures.RiskAssessment.InsurancePremium
- ProtectionMeasures.RiskAssessment.RiskRating
- ProtectionMeasures.ResilienceMeasures.FloodGates
- ProtectionMeasures.ResilienceMeasures.FloodBarriers
- ProtectionMeasures.ResilienceMeasures.SumpPump
- ProtectionMeasures.ResilienceMeasures.FloodWarningSystem

In the flat mapping, these appear as:^[34]

- flood_zone
- overall_flood_risk
- flood_risk_type
- ground_level_meters
- elevation (alias of ground_level_meters)
- river_distance
- insurance_premium
- risk_rating
- flood_gates
- flood_barriers

- sump_pump
- flood_warning_system

These fields, together with basic location and ID, are what you would feed into any BRI or resilience index model even though the code doesn't label them "BRI".^[34] ^[35]

Would you like me to propose a concrete "BRI field set" (e.g. required + optional) based on these, so you can wire it into your validator or UI as a named group?



OK review the BRI analysis from previous chat and outline your proposed fields

Based on the BRI-relevant analysis you're doing on top of the Property CDM, the proposed BRI field set is essentially the risk, resilience, and core identification/location fields needed to calculate a property-level resilience index.

Below I'll split them into core required and recommended optional fields, and show both nested and flat names.

1. Core required BRI fields

These are the fields I'd treat as required for BRI calculations (beyond your current 4 CDM required fields). They're either already required in the validator or are clearly essential for any flood-/risk-based index.^[45] ^[46]

Identification and location

Already required in current CDM:

- PropertyHeader.Header.PropertyID
- PropertyHeader.Header.CatchmentID
- PropertyHeader.Location.LatitudeDegrees
- PropertyHeader.Location.LongitudeDegrees

Flat mapping equivalents:^[47]

- property_id
- catchment_id
- latitude
- longitude

Exposure / hazard

From `PropertyHeader.RiskAssessment`:^[46]

- `PropertyHeader.RiskAssessment.EAFloodZone`
- `PropertyHeader.RiskAssessment.OverallFloodRisk`
- `PropertyHeader.RiskAssessment.FloodRiskType`
- `PropertyHeader.RiskAssessment.GroundLevelMeters`

Flat keys:^[47]

- `flood_zone`
- `overall_flood_risk`
- `flood_risk_type`
- `ground_level_meters`
- `elevation` (alias; same value as `ground_level_meters`)

I would treat `flood_zone`, `overall_flood_risk`, and `ground_level_meters` as required for a BRI score, since they give you “where it is”, “how risky”, and “how high above sea level”.^[46] ^[47]

Basic protection / resilience

From `ProtectionMeasures.RiskAssessment` and `ProtectionMeasures.ResilienceMeasures`:^[46]

- `ProtectionMeasures.RiskAssessment.RiskRating`
- `ProtectionMeasures.ResilienceMeasures.FloodGates`
- `ProtectionMeasures.ResilienceMeasures.FloodBarriers`

Flat keys:^[47]

- `risk_rating`
- `flood_gates`
- `flood_barriers`

I'd mark `risk_rating` as required for BRI if you intend to use a single composite risk indicator, and at least one of `flood_gates` / `flood_barriers` as essential for physical protection scoring.^[46] ^[47]

So a minimal BRI required set could be:

- `property_id`
- `catchment_id`
- `latitude`
- `longitude`
- `flood_zone`
- `overall_flood_risk`
- `ground_level_meters` (and hence elevation)

- risk_rating
- flood_gates
- flood_barriers

(plus their nested equivalents).

2. Recommended optional BRI fields

These add depth to the BRI analysis (better severity, vulnerability, and financial impact modelling) but don't have to be strictly required.

Additional location / proximity

- PropertyHeader.RiskAssessment.RiverDistanceMeters → river_distance [\[47\]](#) [\[46\]](#)

Broader resilience measures

- ProtectionMeasures.ResilienceMeasures.SumpPump → sump_pump
- ProtectionMeasures.ResilienceMeasures.FloodWarningSystem → flood_warning_system [\[46\]](#) [\[47\]](#)

Financial risk / impact

- ProtectionMeasures.RiskAssessment.InsurancePremium → insurance_premium [\[47\]](#) [\[46\]](#)

These are excellent BRI enrichers: river distance refines hazard; sump pump and warning systems improve resilience scoring; insurance premium gives a market-based view of perceived risk. [\[46\]](#) [\[47\]](#)

3. Table: proposed BRI field group

BRI role	Nested CDM field	Flat key	Status	
Identifier	PropertyHeader.Header.PropertyID	property_id	Required	
Catchment	PropertyHeader.Header.CatchmentID	catchment_id	Required	
Latitude	PropertyHeader.Location.LatitudeDegrees	latitude	Required	
Longitude	PropertyHeader.Location.LongitudeDegrees	longitude	Required	
Flood zone	PropertyHeader.RiskAssessment.EAFloodZone	flood_zone	Required	
Overall risk	PropertyHeader.RiskAssessment.OverallFloodRisk	overall_flood_risk	Required	
Flood risk type	PropertyHeader.RiskAssessment.FloodRiskType	flood_risk_type	Optional	
Elevation	PropertyHeader.RiskAssessment.GroundLevelMeters	ground_level_meters / elevation	Required	
River distance	PropertyHeader.RiskAssessment.RiverDistanceMeters	river_distance	Optional	

BRI role	Nested CDM field	Flat key	Status	
Composite rating	ProtectionMeasures.RiskAssessment.RiskRating	risk_rating	Required	
Premium	ProtectionMeasures.RiskAssessment.InsurancePremium	insurance_premium	Optional	
Flood gates	ProtectionMeasures.ResilienceMeasures.FloodGates	flood_gates	Required	
Flood barriers	ProtectionMeasures.ResilienceMeasures.FloodBarriers	flood_barriers	Required	
Sump pump	ProtectionMeasures.ResilienceMeasures.SumpPump	sump_pump	Optional	
Warning system	ProtectionMeasures.ResilienceMeasures.FloodWarningSystem	flood_warning_system	Optional	[47] [46]

If you like, next step I can:

- Sketch how to add a BRI section to PROPERTY_SCHEMA (or a logical group) and
- Update `get_required_fields()` to enforce this BRI required set behind a flag (e.g. `mode="bri"`).

Would you want the validator to always enforce the BRI required fields, or only when a “BRI mode”/flag is explicitly enabled?

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this information

Purpose and capability scope

BRI is a hazard-mapping and resilience assessment tool for the building sector, designed to mainstream climate- and disaster-resilient construction in emerging markets.

It evaluates location-specific climate and natural hazard risks (e.g. flood, windstorm, seismic, heat) for a given real-estate project and assesses whether appropriate mitigation measures are present.

The platform is intended to support developers, investors, lenders, home-buyers and policymakers by: assessing resilience, identifying gaps, guiding upgrades, and disclosing an auditable rating.

It is explicitly positioned as an adaptation complement to EDGE, which covers energy and environmental footprint; EDGE handles operational sustainability, BRI handles physical climate and disaster resilience.

Illustratively, in Vietnam and other pilots BRI is used to flag buildings in high-risk locations, guide the addition of flood defenses or structural strengthening, and provide lenders with evidence that climate risks have been mitigated.

Conceptual assessment framework

From public material you can infer a three-layer logic:

Hazard and exposure layer

Uses web-based hazard mapping to attach local hazard intensities or categories to each project, based on its coordinates and context.

Covers multiple perils (climate-related and geophysical) and recognizes that risk is location-specific, so the same building typology will be scored differently in different places.

Vulnerability and mitigation layer

Collects structured data on building characteristics and resilience measures: structural system, envelope, elevation, drainage, backup power, emergency access, etc.

Looks across the building life-cycle (design, construction, retrofits) and is applicable both to new builds and renovation projects.

The assessment criteria and checklists are standardized, enabling comparable scoring across projects and markets.

Rating and disclosure layer

Translates the hazard–vulnerability–mitigation profile into a resilience rating scale that acts as common language for the property and finance sectors.

The rating is designed to be transparent and disclosable in transactions and portfolios (e.g. for loan covenants, green/resilient bond frameworks, and climate-risk reporting).

Third-party verification is enabled: Bureau Veritas has been appointed as a global verifier to review project data and confirm ratings in key emerging markets.

From a capability perspective, BRI therefore sits at the intersection of hazard analytics, asset-level engineering checks, and ESG/TCFD-style disclosure.

Data and scoring structure

IFC does not publish the full schema in the same way that, say, buildingSMART publishes the IFC EXPRESS model, but public guides and webinars make the main structural features reasonably clear.

Core data objects

At minimum, a BRI assessment organizes data into:

Project/asset entity

Identified building or development (location, type, size, use, ownership, stage).

Gross floor area is tracked because IFC uses it in reporting uptake (e.g. millions of m² assessed).

Hazard profile

For each relevant peril, the platform stores a hazard category or intensity band for the site, derived from linked hazard layers or external providers.

This is effectively a set of location-keyed attributes: hazard type, severity class, return-period band, and possibly climate-change adjustment.

Resilience measures / attributes

A structured checklist of building-level measures, grouped by systems (structure, envelope, MEP, site/landscape, emergency systems, operations).

Each measure is recorded as present/absent and in some cases qualified (e.g. design standard used, elevation above flood level, redundancy level).

Assessment outcome and rating

A composite resilience score and categorical rating (e.g. tiers that reflect adequacy of measures relative to hazards).

Supplementary outputs: narrative explanation and a set of indicators that can be fed into project finance documentation or green/resilient taxonomies.

Computation and logic

Public descriptions point to the following logic rather than raw equations:

For each hazard, BRI tests whether minimum and advanced measures appropriate to that hazard and hazard level are in place.

The tool aggregates these into dimension-level scores (e.g. site planning, structural robustness,

building services continuity, emergency preparedness) and then into an overall rating. The rating is context-normalized: a building in a low-hazard area does not need the same level of defenses as one in an extreme-hazard zone to reach the same category, but both must be demonstrably resilient relative to their exposure.

Because BRI targets finance and disclosure as much as engineering, there is emphasis on binary and categorical fields that can be easily audited and aggregated, rather than continuous structural parameters or fragility curves.

Relation to other standards and models

BRI is part of IFC's broader sustainable buildings toolkit, alongside EDGE and the Sustainable Buildings Finance Reference Guide, and is referenced explicitly in the latter as the building-level resilience tool.

It is designed to plug into green and sustainability finance, supporting banks that want to differentiate loans or investments for climate-resilient buildings and respond to investor pressure on physical climate risk.

In the BIM world, this is distinct from buildingSMART's IFC data model (Industry Foundation Classes): IFC defines an open schema and object relationships for BIM, whereas BRI defines a domain-specific questionnaire, hazard linkages, and rating algorithm on top of generic building and location data.

Conceptually, you could represent a BRI assessment as an IFC-aligned property set (e.g. a Pset_BRIResilience attached to an IfcBuilding), but that mapping is not formally standardized at present.

Given that BRI is a hazard- and resilience-rating tool layered on top of generic building/location data, the Property CDM needs a set of fields that mirror BRI's own structure: project/asset, multi-hazard profile, resilience measures by system, and rating/disclosure outputs.^[58] ^[59]

Below is a proposed field set, grouped to align with the BRI description you pasted.

1. Project / asset entity fields

These give BRI the "what building, where, how big, used for what" layer.

You already have many of these in CDM; I'm just grouping and naming them explicitly as your BRI Project object.^[59]

Identification and location

- PropertyHeader.Header.PropertyID
- PropertyHeader.Header.CatchmentID
- PropertyHeader.Location.BuildingNumber
- PropertyHeader.Location.StreetName
- PropertyHeader.Location.TownCity
- PropertyHeader.Location.County
- PropertyHeader.Location.Postcode
- PropertyHeader.Location.LocalAuthority
- PropertyHeader.Location.Country

- PropertyHeader.Location.Region
- PropertyHeader.Location.UrbanRuralClassification
- PropertyHeader.Location.LatitudeDegrees
- PropertyHeader.Location.LongitudeDegrees

Flat equivalents: `property_id`, `catchment_id`, `building_number`, `street_name`, `town_city`, `county`, `postcode`, `local_authority`, `country`, `region`, `urban_rural`, `latitude`, `longitude`.^[58]

Type, size, use, stage

- PropertyHeader.Header.propertyType
- PropertyHeader.PropertyAttributes.OccupancyType
- PropertyHeader.PropertyAttributes.PropertyResi
- PropertyHeader.PropertyAttributes.PropertyAreaSqm
- PropertyHeader.PropertyAttributes.NumberOfStoreys
- PropertyHeader.PropertyAttributes.ConstructionYear
- PropertyHeader.PropertyAttributes.PropertyPeriod
- PropertyHeader.PropertyAttributes.PropertyCondition

These give you “building type, use, size, age, condition” that BRI uses to interpret vulnerability and measure adequacy.^[59]

2. Hazard profile fields (multi-hazard)

BRI attaches hazard categories/intensities by peril to the project’s coordinates. Your schema already has a flood-centric start; I’d formalize that as a **hazard profile** block.^[59]

Existing Property CDM hazard fields

- PropertyHeader.RiskAssessment.EAFloodZone → site flood zone / hazard band
- PropertyHeader.RiskAssessment.OverallFloodRisk → summarized flood risk level
- PropertyHeader.RiskAssessment.FloodRiskType → main flood peril(s)
- PropertyHeader.RiskAssessment.GroundLevelMeters → absolute elevation (used as proxy for hazard/exposure)
- PropertyHeader.RiskAssessment.RiverDistanceMeters → distance to main river / water body

Flat equivalents: `flood_zone`, `overall_flood_risk`, `flood_risk_type`, `ground_level_meters` / `elevation`, `river_distance`.^[58]

BRI-aligned additional hazard slots (conceptual)

Based on the BRI description (flood, windstorm, seismic, heat, etc.), in your CDM you’d reserve slots such as:

- HazardProfile.Flood.HazardCategory
- HazardProfile.Flood.ReturnPeriodBand

- HazardProfile.Flood.ClimateAdjustmentFlag
- HazardProfile.Windstorm.HazardCategory
- HazardProfile.Seismic.HazardCategory
- HazardProfile.Heat.HazardCategory

These don't yet exist in the current code, but conceptually they are the "hazard type, severity class, return-period band, climate-change adjustment" attributes mentioned in the BRI text. You'd either extend `PropertyHeader.RiskAssessment` or add a new `HazardProfile` section to the schema to hold them.^[59]

3. Resilience measures / building attributes by system

BRI collects a structured checklist by system: structure, envelope, MEP, site/landscape, emergency and operations. Your CDM currently has mainly exposure and flood-resilience measures, so these are the BRI-specific measures you can support now plus the obvious extension points.

Already present and clearly BRI-relevant

From `ProtectionMeasures.ResilienceMeasures`:^[59]

- ProtectionMeasures.ResilienceMeasures.FloodGates
- ProtectionMeasures.ResilienceMeasures.FloodBarriers
- ProtectionMeasures.ResilienceMeasures.SumpPump
- ProtectionMeasures.ResilienceMeasures.FloodWarningSystem

Flat: `flood_gates, flood_barriers, sump_pump, flood_warning_system`.^[58]

From `PropertyHeader.Construction` and `PropertyHeader.PropertyAttributes` (good proxies for structural and envelope characteristics):^[59]

- PropertyHeader.Construction.ConstructionType
- PropertyHeader.Construction.FoundationType
- PropertyHeader.Construction.FloorLevelMeters
- PropertyHeader.Construction.BasementPresent
- PropertyHeader.PropertyAttributes.NumberOfStoreys
- PropertyHeader.PropertyAttributes.PropertyCondition
- PropertyHeader.PropertyAttributes.ConstructionYear
- PropertyHeader.PropertyAttributes.PropertyPeriod

These map naturally to BRI's structural and site/elevation dimensions (elevation above ground, presence of basement, structural system, age and condition).

Recommended BRI-measure groupings in your CDM

If you logically group what you already have into BRI systems, you get:

- Structure
 - ConstructionType
 - FoundationType
 - NumberOfStoreys
 - BasementPresent
- Site / elevation / access
 - GroundLevelMeters
 - FloorLevelMeters
 - RiverDistanceMeters
- Flood protection
 - FloodGates
 - FloodBarriers
 - SumpPump
- Preparedness / operations
 - FloodWarningSystem
 - (later: backup power, emergency access, etc. if you add them)

This grouping matches the “checklist by systems and life-cycle” emphasis in the BRI description, even though your current schema doesn’t yet cover wind, seismic, or all operational systems.^[59]

4. Rating and disclosure fields

BRI converts hazard + measures into an auditable rating, with narrative and indicators for finance and disclosure. In your code, you already have a generic risk rating and financial proxy.

Current CDM fields you can use for BRI rating layer

- ProtectionMeasures.RiskAssessment.RiskRating → composite risk/resilience level
- ProtectionMeasures.RiskAssessment.InsurancePremium → financial reflection of risk

Flat: risk_rating, insurance_premium.^[58]

On top of that, a BRI-aligned extension to the CDM (in your own schema, not yet in code) would include:

- BRI.Rating.ScoreNumeric
- BRI.Rating.Category (e.g. Bronze/Silver/Gold or A/B/C)
- BRI.Rating.VerifiedBy (e.g. Bureau Veritas / other verifier)
- BRI.Rating.VerificationDate
- BRI.Rating.NarrativeSummary
- BRI.Rating.DisclosureFlags (e.g. allowed in loan docs, ESG reports)

These match the “resilience rating scale, disclosable, third-party verification” aspects described in the BRI text, even though they aren’t yet present in your Property CDM code. You’d implement them as an additional top-level BRI section or embed them under `ProtectionMeasures` or a new `AssessmentOutcome` section.^[59]

5. Suggested minimum BRI field subset (from your CDM)

Pulling the above into a concise, practical subset you can actually enforce today without extending the schema:

Required core

- `property_id`, `catchment_id`, `latitude`, `longitude`
- `flood_zone`, `overall_flood_risk`, `flood_risk_type`
- `ground_level_meters` / `elevation`
- `construction_type`, `foundation_type`, `number_storeys`, `basement_present`
- `flood_gates`, `flood_barriers`
- `risk_rating`

Strongly recommended

- `property_area_sqm`, `occupancy_type`, `property_resi`
- `construction_year`, `property_period`, `property_condition`
- `river_distance`
- `sump_pump`, `flood_warning_system`
- `insurance_premium`

All of these already exist in `PROPERTY_SCHEMA` and are mapped by `create_mapping`, so you can turn them into a named “BRI field group” in your UI or validator without changing the underlying code structure—only the validation rules and documentation.^{[58] [59]}

Do you want me to turn this into a concrete schema addition, for example a BRI section in `PROPERTY_SCHEMA` or a JSON/YAML spec you can hand to front-end/DB teams?

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What is the BRI methodology that leads them to define their resilience category

BRI defines a building’s resilience category by combining multi-hazard exposure with a structured checklist of resilience measures, then applying a “weakest-link” ruleset to derive a letter-grade rating (NR/B/A/AA, sometimes with a “+” for operational continuity).^{[60] [61] [62] [63] [64]}

Below is the methodology in practical terms.

Overall approach

- BRI is a web-based questionnaire and hazard-mapping tool: it links a building's geolocation to hazard layers and then asks targeted questions about design, construction and operations. [\[62\]](#) [\[63\]](#) [\[64\]](#) [\[60\]](#)
- It produces a standard letter-grade resilience rating that can be disclosed to lenders, insurers, investors and regulators as an auditable indicator of physical climate and disaster resilience. [\[63\]](#) [\[60\]](#) [\[62\]](#)

Step 1 – Hazard and exposure profiling

- For the project's coordinates, BRI pulls in location-specific hazard information from web-based hazard maps (e.g. flood, windstorm, seismic, heat and other perils). [\[64\]](#) [\[60\]](#) [\[62\]](#) [\[63\]](#)
- For each applicable hazard, it assigns a hazard band or category (intensity class, return-period band, possibly with a climate-change adjustment), so hazard is represented as a set of categorical attributes per peril. [\[61\]](#) [\[62\]](#) [\[64\]](#)

This defines what the building needs to be resilient *to* in order to achieve a high rating.

Step 2 – Checklist of resilience measures

- The core of the tool is a standardized questionnaire that captures whether specific resilience measures are present, absent or partially met. [\[62\]](#) [\[63\]](#) [\[64\]](#)
- Measures are grouped by systems or dimensions, such as structure, envelope, MEP/services, flood and drainage, emergency systems, site/landscape and operational continuity. [\[61\]](#) [\[63\]](#) [\[64\]](#)
- For each hazard and system, BRI defines minimum and advanced measures appropriate to that hazard level (for example, flood defenses and elevated critical equipment in high flood-risk zones, structural detailing for seismic areas, wind resistance for cyclones, etc.). [\[63\]](#) [\[64\]](#) [\[61\]](#)

The emphasis is on binary/categorical fields (yes/no, compliant/non-compliant, level met) that can be easily audited and compared across projects. [\[64\]](#) [\[61\]](#) [\[62\]](#)

Step 3 – Dimension-level scoring

- Responses for individual measures are aggregated into dimension-level scores, typically covering themes such as site planning, structural robustness, building services continuity and emergency preparedness. [\[61\]](#) [\[63\]](#) [\[64\]](#)
- Within each dimension, the scoring tests whether all required measures for each applicable hazard are met, and whether any “advanced” measures are in place, often using simple rule-based logic instead of continuous engineering models. [\[63\]](#) [\[64\]](#) [\[61\]](#)

Each dimension's score effectively captures “how well this building is prepared for its hazard profile in that aspect”.

Step 4 – Weakest-link aggregation and context-normalization

- The overall BRI rating uses a **weakest-link principle**: the risk for all applicable local hazards must be adequately mitigated; serious gaps for any one key hazard will cap the overall rating. [\[62\]](#) [\[64\]](#) [\[61\]](#)
- Ratings are **context-normalized**: a building in an extreme-hazard area must meet more stringent measure sets to achieve the same category than an equivalent building in a low-hazard area, but both must be demonstrably resilient relative to their exposure. [\[64\]](#) [\[61\]](#) [\[63\]](#)

So the methodology doesn't average good and bad dimensions; instead it ensures there are no critical weaknesses for any hazard that matters at that site.

Step 5 – Rating levels and “plus” modifier

- The brochure describes four main levels (NR, B, A, AA), interpreted as: [\[60\]](#)
 - NR – fails to incorporate most recommended practices; unlikely to withstand most applicable hazards even at moderate levels.
 - B – incorporates some recommended practices; likely to withstand applicable hazards at moderate level.
 - A – incorporates most recommended practices; likely to withstand applicable hazards at moderate-high level.
 - AA – incorporates all recommended practices for all applicable hazards, typically above local standards; likely to withstand hazards at high level.
- A “+” suffix indicates that, in addition to physical measures, operational continuity requirements are also met (e.g. backup systems, emergency planning), so e.g. A+. [\[60\]](#) [\[61\]](#)

The rating is therefore a structured summary of how many recommended measures are present relative to what's needed for the local hazard context.

Step 6 – Verification and disclosure

- The methodology is designed for use in finance and disclosure, so ratings are meant to be transparent, traceable to input fields, and suitable for verification. [\[60\]](#) [\[62\]](#) [\[63\]](#)
- IFC has appointed independent verifiers (e.g. Bureau Veritas) to review data and confirm ratings in key markets, turning the rating into a credible signal for banks, insurers and investors. [\[63\]](#) [\[60\]](#)

Put together, BRI's resilience category is:

1. hazard-informed, 2) measure-driven via a structured checklist, 3) aggregated using weakest-link logic across hazards and dimensions, and 4) expressed as a letter grade with optional “+” for continuity, suitable for third-party verification and disclosure. [\[61\]](#) [\[62\]](#) [\[64\]](#) [\[60\]](#) [\[63\]](#)

If helpful, I can next translate this into a concrete scoring outline (pseudo-code or rule table) that you could implement on top of your Property CDM fields.

can you list the components in the standardised checklist

The BRI standardized checklist is organised by hazard category (wind, water, fire, geoseismic) and by two big themes: physical integrity and operational continuity. Each theme has groups of measures, and each group has specific “yes/no/level” checklist items.^{[68] [69]}

Below I'll list the components at the group/measure-set level (not every individual question line), which is how you'd mirror it in your CDM/BRI model.

Top-level components

Across the whole tool, the standardized checklist is structured as:^{[69] [68]}

- Hazard applicability and level (per hazard type)
- Risk-reduction measures for **Physical Integrity**
- Risk-reduction measures for **Operational Continuity**

Each item is a discrete measure that can be checked off by the Assessor and later verified (two verifiers must confirm each measure).^[68]

Physical integrity – hazard categories and measure groups

The User Guide splits physical-integrity measures into four hazard categories: Wind, Water, Fire, Geoseismic. Each hazard section has its own checklist, but the items fall into recurring groups such as site, structure, envelope, services and critical equipment.^[68]

1. Wind (storms, tornados, downbursts)

Components of the wind checklist include groups of measures like:^[68]

- Site and exposure
 - Siting relative to terrain and topography
 - Avoidance or treatment of highly exposed configurations
- Structural system and design standard
 - Main wind-resisting system sized to specified design wind speeds (e.g. up to AA threshold ~290 km/h for storms/tornados, 200 km/h for default downbursts)
 - Anchorage and load paths from roof to foundations
 - Compliance with or going beyond local wind-design codes / global best practice (e.g. South Florida High Velocity Building Code)
- Building envelope and openings
 - Wind-rated roofing (materials, fastening, edge details)

- Wind-resistant cladding and connections
- Protected/impact-resistant windows, doors, shutters
- Garage and large openings reinforced for wind loads
- Rooftop and exterior equipment
 - Secure anchoring of rooftop plant and equipment
 - Protection/anchoring for façade-mounted elements and signage
- Nonstructural elements and debris
 - Measures to reduce wind-borne debris (design and housekeeping)

Each of these groups is implemented as a set of checklist measures that must be satisfied at B/A/AA levels, depending on local wind hazard.^[68]

2. Water (flood, storm surge, tsunami, local/urban flood)

Water-related checklists cover:^[69] ^[68]

- Site selection and topography
 - Siting relative to floodplains, water bodies, natural flow paths
 - Consideration of sea-level rise, storm surge or tsunamizones where relevant
- Elevation and structural measures
 - Minimum finished floor elevation above design flood level
 - Use of stilts/piles or raised platforms in high-risk areas
 - Dry flood-proofing or wet flood-proofing strategies as appropriate
- Drainage and stormwater management
 - On-site drainage design (surface drainage, swales, channels)
 - Permeable surfaces and reduced runoff
 - Connection to and capacity of surrounding drainage systems
- Flood barriers and defenses
 - Permanent or deployable flood barriers/gates around entrances and vulnerable areas
 - Backflow prevention and protection of drainage inlets/outfalls
- Protection of critical equipment and services
 - Elevation or protection of electrical switchgear, pumps, HVAC, fuel storage
 - Non-return valves, sump pumps and emergency drainage arrangements

These measures are checked against the mapped water hazard class (local/urban, river, coastal, flash, storm surge, tsunami) for the specific site.^[68]

3. Fire (local fire, wildfire)

The fire checklist is grouped into:^[68]

- Structural fire resistance and compartmentation
- Fire-resistant materials and finishes
- Fire detection and suppression systems (alarms, sprinklers, hydrants)
- Wildfire zone-specific measures (defensible space, non-combustible zones near buildings)

BRI assumes life-safety emergency provisions are largely in the building code; its focus is whether asset-protection and continuity-oriented measures are present.^[68]

4. Geoseismic (earthquake, volcano, landslides, subsidence)

The geoseismic checklist covers:^[68]

- Site and geotechnical conditions
 - Siting relative to known faults, landslide-prone slopes, volcanic hazards
 - Ground investigation and geotechnical design basis
- Seismic structural design and detailing
 - Building designed to or above appropriate seismic code level
 - Ductile detailing, confinement, and regularity of structural system
- Foundations and soil–structure interaction
 - Foundation type suitable for seismic/landslide/subsidence risk
 - Measures to mitigate liquefaction or differential settlement where relevant
- Nonstructural seismic measures
 - Anchorage of non-structural components (ceilings, partitions, heavy equipment)
 - Restraint of building contents that can cause injury or damage

Again, each of these is expressed as specific checkboxes in the BRI app.^[68]

Operational continuity – checklist components for “+”

Operational continuity measures are a separate block; satisfying a minimum number on top of a non-NR resilience rating yields the “+” modifier (e.g. A+).^[68]

Groups of continuity measures include:^[68]

- Power supply continuity
 - Backup generators and fuel storage
 - Redundant feeds or automatic transfer switches
- Water supply and sanitation continuity
 - Backup water storage and pumping
 - Protection of water treatment and sewage systems

- Telecoms and critical IT
 - Protection/redundancy for communications infrastructure and critical IT equipment
- Access and egress under hazard conditions
 - Access routes designed to remain usable or to recover quickly after events
 - Safe access/egress points relative to likely hazard impacts (e.g. flooding)
- Business / operations continuity planning
 - Documented continuity and emergency plans
 - Procedures for maintaining essential functions during and after events

The User Guide specifies that at least a defined minimum number of these operational continuity measures must be met for the “+” designation.^[68]

How this maps to a “standard checklist”

In practice, the standardized checklist is:

- A set of hazard-applicability and intensity questions.
- Four hazard-category sections (Wind/Water/Fire/Geoseismic), each with grouped measures for site, structure, envelope, services, equipment and nonstructural elements.
- A separate Operational Continuity section with grouped measures for power, water, telecoms, access, and continuity planning.^{[69] [68]}

If you’d like, I can next draft a condensed “BRI-style checklist” table tailored to your Property CDM fields (e.g. which of your existing fields would feed each group, and what new boolean fields you’d add).

✱

the checklist. checklist as in I want to know what checks are on the checklist not who is doing the checking.

The BRI checklist is essentially a list of mitigation measures grouped by hazard and by whether they protect **physical integrity** or **operational continuity**. Below I’m focusing only on *what* is checked (the measures), not on who checks them.^{[77] [78] [79]}

Because IFC does not publish every single question line publicly, what we can see is the set of measure categories and typical checks that appear in the official brochure, notice and user-guide descriptions.^{[78] [80] [77]}

Wind hazard checklist – example measures

For windstorms, cyclones, downbursts, the checklist covers:^[80] ^[77]

- Site and exposure
 - Building sited and oriented to reduce wind exposure (terrain/topography considered).
- Structural wind resistance
 - Main wind-resisting system designed for mapped design wind speeds for the site.
 - Continuous load path from roof to foundation.
- Roof and envelope
 - Wind-rated roof system (materials, fixings, edge details) installed.
 - External cladding, windows and doors rated/anchored for design wind loads.
- Openings and large doors
 - Large doors (e.g. garages, loading bays) strengthened or braced for wind.
- Equipment and attachments
 - Rooftop and façade-mounted equipment anchored for design wind speeds.

These appear in the App as individual “Is this measure present / compliant?” checks.^[79] ^[77]

Water/flood hazard checklist – example measures

For river/coastal/flash/urban flooding and storm surge, checks include:^[77] ^[78] ^[80]

- Site and flood zone
 - Project avoids highest-risk flood zones where feasible.
- Elevation and structure
 - Finished floor elevation above design flood level.
 - Elevated or flood-resistant design (stilts, raised plinth, stilts/piles) where required.
- Flood entry and barriers
 - Permanent or deployable barriers/gates for doors, windows, and other openings in flood-exposed areas.
 - Backflow prevention devices on drains / sewers.
- Drainage and runoff
 - On-site drainage sized for design storm, with positive flow paths.
 - Use of permeable surfaces or retention to reduce runoff.
- Protection of critical systems
 - Electrical, mechanical and other critical plant elevated above flood levels or protected by enclosures.
 - Sump pumps or equivalent de-watering systems installed for basements or low points.

Again, each of these is an item or small set of items in the flood section of the checklist.^[78] ^[79] ^[77]

Fire hazard checklist – example measures

For building fire and wildfire, typical checklist components are:^[80] ^[77]

- Structural fire resistance
 - Structural frame and key elements designed to required fire-resistance ratings.
- Compartmentation and materials
 - Fire-resistant walls, doors and penetration seals between key zones.
 - Non-combustible or fire-resistant materials in critical areas.
- Detection and suppression
 - Automatic fire detection and alarm systems installed and functional.
 - Fixed suppression (e.g. sprinklers, hydrants, hose reels) installed per hazard and building use.
- Wildfire-specific measures (where applicable)
 - Defensible space and vegetation management around buildings.
 - Non-combustible zone adjacent to external walls/roofs.

Each is recorded as present/absent and, where relevant, whether it meets B/A/AA thresholds.^[77]

Geoseismic hazard checklist – example measures

For earthquake, landslide, subsidence and volcanic hazards, the checklist includes:^[77]

- Site and ground conditions
 - Site investigated for seismic/landslide/subsidence risk and suitable for proposed use.
- Seismic structural design
 - Building designed to at least code-required seismic level for the site (or higher for top ratings).
 - Ductile detailing and regular structural layout.
- Foundations and soil mitigation
 - Foundation type selected to manage expected seismic/ground movements.
 - Specific measures for liquefaction or unstable slopes where relevant.
- Non-structural seismic protection
 - Anchoring/bracing of ceilings, partitions and other non-structural components.
 - Anchorage of heavy equipment and building services.

These are again binary/categorical checks in the geoseismic section.^[78] ^[77]

Operational continuity checklist – example measures for “+”

For the “+” modifier (continuity), BRI has separate checks, such as:^[77]

- Power continuity
 - Backup generator(s) with adequate capacity and protected location.
 - Fuel storage and refuelling arrangements.
- Water and sanitation continuity
 - Backup water storage or alternative supply for essential functions.
 - Protection of pumps, treatment and sewage systems.
- Telecommunications / IT
 - Redundant or protected communications links and key IT hardware.
- Access and logistics
 - At least one access route designed to remain usable or recover quickly after hazards.
- Business/operations continuity
 - Documented continuity/emergency plans aligned with identified hazards.

The User Guide then defines how many of these continuity measures must be ticked, in addition to the physical-integrity measures, to get the “+”.^[77]

In short, the BRI “standardised checklist” is a structured list of mitigation measures across:

- Wind, water, fire, geoseismic physical-integrity measures, and
- Operational continuity measures (power, water, IT, access, planning),

each expressed as specific yes/no (or level) checks against the hazard context of the site.^{[79] [80] [78] [77]}

For your Property CDM, do you want to mirror this as separate boolean fields per measure (e.g. `has_backup_generator`, `roof_wind_rated`), or start with higher-level grouped flags (e.g. `wind_structure_measures_met`, `flood_measures_met`)?

✱

So no checklist, then how can anyone improve their rating if they don't know the questions?

You *can* improve your BRI rating, but the detailed question-by-question checklist is not fully published in a simple web page; it lives inside the BRI web/app interface and in long user-guide PDFs that IFC distributes (100–200 pages). Public docs describe the structure (hazard categories and measure groups) and many example measures, but they don't dump a short, human-readable list of every single yes/no question.^{[92] [93] [94]}

Here's what that means in practice:

1. Where the actual questions live

- BRI is implemented as an online questionnaire: when you start a project, the app pulls hazard data for your location and then shows you a tailored set of questions about design, construction and operations.^{[95] [96] [97]}
- The full “checklist” is essentially the form definition inside that app (and its admin configuration), not a one-page PDF checklist.^{[96] [94]}
- IFC’s user guides describe the structure, logic and many example measures, but they are long technical manuals rather than a short bullet list of every single item.^{[93] [94] [92]}

So: there *is* a checklist; it’s just embedded in the digital tool and documentation, not published as a simple, public one-pager.

2. What you *do* get publicly

The public material gives you:

- The four hazard categories (wind, water, fire, geoseismic) and operational continuity as the main blocks.^{[97] [98] [95]}
- The main measure groups within each: site selection, structural system, elevation, flood defenses, drainage, fire protection, seismic detailing, protection of critical equipment, backup power, water, access, etc.^{[94] [95] [93]}
- Descriptions of minimum vs advanced measures for B, A, AA ratings, with examples (e.g. elevation above design flood level, code-plus seismic design, anchored rooftop equipment, backup generator).^{[98] [99] [95]}

That’s enough to understand what to improve (add elevation, flood barriers, backup power, etc.), even if you don’t see every individual checkbox label.^{[95] [98]}

3. How users actually improve their rating

In real use, people improve BRI ratings by:^{[96] [97] [95]}

- Creating or opening a project in the BRI app, answering all hazard and measure questions, and seeing which measures are missing for higher levels.
- The app highlights gaps against the target rating (e.g. you’re at B, to reach A you need specific additional measures for the hazards at your site).
- They then update designs (e.g. raise critical equipment, add flood barriers, upgrade wind resistance) and re-run the assessment.

Because it’s a self-declaration tool, the platform itself is the primary “checklist UI”.

4. What you can realistically mirror in your CDM

For your Property CDM and your own BRI-style modelling, you don’t need IFC’s exact wording for every item. You can:

- Take the published measure groups and examples (e.g. “finished floor above design flood level”, “backup generator”, “anchored rooftop equipment”) and implement them as boolean / categorical fields in your schema.^{[98] [94] [95]}

- Define your own explicit checklist questions that are functionally equivalent (same engineering intent) even if phrased differently.
- Use a similar weakest-link scoring logic to derive a rating.^{[94] [98]}

That way, your users *will* see a clear list of checks they can improve, even if the official IFC checklist is not fully public.

If you want, I can take the BRI docs as a basis and draft a concrete, explicit checklist for your system (e.g. 20–40 yes/no questions mapped onto your Property CDM fields) so your users have a transparent “to-do list” to raise their resilience score.



yes I want an explicit checklist so that all of them are included in the CDM

You can absolutely have an explicit checklist in *your* CDM; we just have to build it from the public BRI structure rather than copy IFC’s internal form verbatim. IFC exposes the logic, hazard categories, and measure types, but not a short, public list of every in-app question, so the way to get a full checklist is:^{[108] [109]}

- Take the publicly described measure groups and examples, and
- Turn them into an explicit set of yes/no (or categorical) checks in your own schema.^{[109] [110] [108]}

I can do that for you.

Before I draft that list, one thing will change how detailed it should be:

Do you want a relatively compact checklist (say 20–30 items) or are you happy with a more granular one (40–60+ items) that maps more directly to BRI’s groups (e.g. separate items for each hazard and system)?



more granular

You can implement a granular, BRI-style checklist in your CDM by turning the public measure groups into explicit yes/no (or categorical) fields. Below is a 50-ish item checklist, grouped so you can drop each item straight into your schema as a boolean or enum.

Each bullet is a *check* you would ask about a given building.

A. Hazard profile (context checks)

These are derived, not measures, but you’ll want explicit fields for them so the rest of the checklist can be hazard-aware.^{[124] [125]}

- Site is in mapped flood hazard class (e.g. low/medium/high/extreme).

- Site is in mapped windstorm/cyclone hazard class.
- Site is in mapped seismic hazard class.
- Site is in mapped wildfire or urban fire-risk class.

(You already partially cover this with EAFloodZone, OverallFloodRisk, FloodRiskType; you'd add similar categorical fields for wind/seismic/heat/wildfire.) ^[126] ^[124]

B. Wind hazard checklist (storms, cyclones, downbursts)

B1. Site and exposure ^[124] ^[126]

- Building siting accounts for wind exposure (terrain/topography assessed).
- Building orientation / massing reduces wind pressure and funneling where practicable.

B2. Structural wind resistance

- Primary structural system is designed for mapped design wind speed for the site (code-compliant or better).
- Continuous vertical load path from roof to foundations is provided for wind loads.
- Lateral bracing / shear systems are detailed to prevent soft-storey or torsional failures under design wind speeds.

B3. Roof system and envelope ^[124]

- Roof covering is rated and fixed for design wind speed (fasteners, edge restraint, detailing).
- Roof edges, parapets and eaves are detailed to prevent wind-uplift failures.
- External cladding is rated and anchored for design wind pressures.

B4. Openings and large doors

- External doors and windows in wind-exposed façades are rated or protected for design wind loads.
- Large doors (garage, loading bay, hangar, etc.) are reinforced/braced or protected against wind failure.

B5. Equipment and attachments

- Rooftop mechanical/electrical equipment is anchored for design wind speeds.
- Façade-mounted elements (signage, louvers, awnings) are anchored to resist design wind loads.

C. Water / flood hazard checklist

(For fluvial, pluvial, coastal, storm-surge and local flooding.) ^[126] ^[124]

C1. Site and floodplain

- Site flood hazard (floodplain, storm surge, tsunami, local flooding) has been explicitly assessed.
- High-risk flood zones are avoided where feasible, or risk is explicitly accepted and mitigated.

C2. Elevation and structural configuration

- Minimum finished floor level of occupied spaces is above design flood level for the site.
- Basement is avoided in high flood-risk areas, or is designed as deliberately floodable with appropriate protection.
- For high flood risk, primary occupied levels are elevated (on stilts, raised plinth or equivalent).

C3. Flood entry, barriers and backflow ^[124]

- Permanent flood-resistant construction is provided at typical entry points (walls, doors, thresholds).
- Deployable flood barriers/gates are provided at all flood-exposed doors and openings below design flood level.
- Backflow prevention devices are installed on sewer/drainage connections to prevent floodwater ingress.

C4. Drainage and stormwater management

- On-site surface drainage is designed for at least the design storm (capacity, slopes, outlets).
- Site layout maintains positive overland flow paths away from buildings.
- Permeable surfaces and/or retention features (e.g. swales, detention basins) are used to manage runoff.

C5. Protection of critical systems ^[124]

- Main electrical switchboards and critical electrical equipment are located above design flood level or protected.
- Main mechanical plant (pumps, boilers, chillers) is located above design flood level or protected.
- Fuel tanks and hazardous material stores are elevated or secured to prevent flotation and leakage during floods.
- Sump pumps or other de-watering systems are installed for basements or low-lying spaces at risk of inundation.

D. Fire / wildfire hazard checklist

D1. Structural fire resistance

- Primary structural frame elements have fire-resistance ratings appropriate to building type and code.
- Fire separation between key zones (e.g. stair cores, plant rooms) is provided to rated levels.

D2. Compartmentation and materials ^[124]

- Fire-rated walls, floors and doors form effective compartments to limit fire spread.
- Penetrations in fire-rated elements are properly sealed with fire-stopping systems.
- External cladding and roof materials in high fire-risk contexts are non-combustible or fire-resistant.

D3. Detection and suppression

- Automatic fire detection and alarm systems are installed and cover all required areas.
- Fire suppression systems (e.g. sprinklers, hydrants, hose reels) are installed per code and functional.

D4. Wildfire-specific measures (where applicable)

- Defensible space (managed vegetation, reduced fuel load) is provided around the building.
- A non-combustible zone is maintained directly adjacent to external walls and roofs (e.g. gravel, paving).

E. Geoseismic hazard checklist

(For earthquake, landslides, subsidence, volcanic hazards.) ^[125] ^[124]

E1. Site and ground conditions

- Site geotechnical and seismic conditions have been formally investigated.
- Site layout avoids known landslide-prone slopes, unstable ground or high subsidence areas where feasible.

E2. Seismic structural design

- Structural system is designed to at least the mapped seismic hazard level for the site.
- Seismic detailing (ductility, confinement, reinforcement) follows a recognized seismic code or standard.
- Structural configuration avoids severe irregularities (e.g. extreme soft storeys, large torsional eccentricity) or mitigates them.

E3. Foundations and soil–structure interaction

- Foundation type is selected and detailed to manage expected seismic and ground movement conditions.
- Where liquefaction is a risk, appropriate ground improvement or foundation design has been implemented.

E4. Non-structural seismic protection

- Suspended ceilings, partitions and heavy fittings are braced/anchored against seismic movement.
- Heavy equipment and building services (e.g. tanks, chillers, switchgear) are anchored and braced.

F. Operational continuity (“+” modifier) checklist

These are the measures BRI uses to award the “+” on top of the base (NR/B/A/AA) rating. ^[126] ^[124]

F1. Power continuity

- Backup power system (generator or equivalent) is installed with capacity to maintain critical functions.
- Backup power equipment and fuel storage are protected from site-specific hazards (flood/wind/fire).

F2. Water and sanitation continuity

- Backup water storage or alternative water supply exists for essential uses.
- Critical pumps and water/sanitation treatment equipment are protected from relevant hazards.

F3. Telecommunications and IT

- At least one backup or redundant telecommunications pathway (e.g. secondary network or protected routing) is provided.
- Critical IT and data infrastructure are protected (e.g. elevated, fire-protected, with power conditioning).

F4. Access and logistics

- At least one access/egress route is expected to remain usable or to be restored quickly under relevant hazards.
- Site access points are designed with hazard impacts in mind (e.g. not solely via flood-prone underpasses).

F5. Business / operations continuity

- A written business/operations continuity plan exists that explicitly addresses local hazard scenarios.
- Emergency response procedures are documented and periodically exercised (drills or tests).

G. How to put this into your CDM

To integrate this into your Property CDM:

- Treat each bullet as a boolean or small enum field (e.g. `wind_structural_design_meets_hazard_code`, `flood_barriers_provided`, `backup_generator_installed`).
- Group them under new sections, for example `BRI.Wind`, `BRI.Flood`, `BRI.Fire`, `BRI.Geoseismic`, `BRI.Continuity`.
- Use your existing CDM fields where they already capture the same concept (e.g. `BasementPresent`, `GroundLevelMeters`, `FloodGates`, `FloodBarriers`, `SumpPump`, `FloodWarningSystem`) and add new fields for items you don't yet represent (backup power, seismic detailing, etc.). [\[127\]](#) [\[128\]](#) [\[124\]](#)

If you want, next I can take this checklist and:

- Map each item explicitly onto your existing CDM fields vs new fields, and
- Sketch a simple scoring rule (pseudo-code) to derive a BRI-style rating from these booleans.

✱

yes map each item explicitly onto existing CDM fields, create an entirely new section that is purely based on the assessment, so there should not have any property header info such as location etc.

You can keep your existing `PropertyHeader` intact and add a new, assessment-only section (for example `BRIAssessment`) that holds all the checklist items as booleans/enums. None of these new fields need to repeat `Location`, `Header` or other identification. [\[129\]](#)

Below I'll:

1. Map where you already have something equivalent in the CDM.
2. Propose a clean `BRIAssessment` structure containing only assessment fields.

1. What already exists in the CDM

Some resilience-related checks are already represented as fields; we just reuse them inside scoring and keep them where they are in the schema.^[130] ^[129]

Existing CDM fields that correspond to checklist concepts

- Flood resilience
 - `ProtectionMeasures.ResilienceMeasures.FloodGates` → flood barrier/defense present
 - `ProtectionMeasures.ResilienceMeasures.FloodBarriers` → additional flood defenses
 - `ProtectionMeasures.ResilienceMeasures.SumpPump` → de-watering / sump pump present
 - `ProtectionMeasures.ResilienceMeasures.FloodWarningSystem` → warning/monitoring present
- Elevation / exposure
 - `PropertyHeader.RiskAssessment.GroundLevelMeters` → absolute elevation
 - `PropertyHeader.Construction.FloorLevelMeters` → floor above ground (helps infer elevation above flood)
- Flood hazard context
 - `PropertyHeader.RiskAssessment.EAFloodZone` → flood hazard band
 - `PropertyHeader.RiskAssessment.OverallFloodRisk`
 - `PropertyHeader.RiskAssessment.FloodRiskType`
 - `PropertyHeader.RiskAssessment.RiverDistanceMeters`
- Structural / building type context
 - `PropertyHeader.Construction.ConstructionType`
 - `PropertyHeader.Construction.FoundationType`
 - `PropertyHeader.Construction.BasementPresent`
 - `PropertyHeader.PropertyAttributes.NumberOfStoreys`
 - `PropertyHeader.PropertyAttributes.PropertyCondition`
 - `PropertyHeader.PropertyAttributes.ConstructionYear`
 - `PropertyHeader.PropertyAttributes.PropertyPeriod`

These are **inputs** to the assessment, not part of the new section; your new `BRIAssessment` will reference them logically but won't duplicate them.^[129]

Everything else in the checklist (wind measures, seismic anchoring, backup power, etc.) needs new fields.

2. New assessment-only section: BRIAssessment

Add a new top-level section alongside PropertyHeader, ProtectionMeasures, TransactionHistory:

```
BRIAssessment
├─ HazardProfile
├─ Wind
├─ Flood
├─ Fire
├─ Geoseismic
└─ Continuity
```

All fields here are pure assessment outputs (checklist items or derived hazard classes), with no location or ID. You can still compute them using existing CDM inputs.

A. BRIAssessment.HazardProfile

These are derived categorizations used by the rest of the checks (even though you may already have some equivalents; here they're standardized for BRI logic).^[131] ^[132]

- `BRIAssessment.HazardProfile.FloodHazardClass` (enum: none / low / medium / high / extreme)
- `BRIAssessment.HazardProfile.WindHazardClass` (enum)
- `BRIAssessment.HazardProfile.SeismicHazardClass` (enum)
- `BRIAssessment.HazardProfile.FireHazardClass` (enum, including wildfire where relevant)

You could derive `FloodHazardClass` from `EAFloodZone` and `OverallFloodRisk` but still store this as a separate assessment field.^[129]

B. BRIAssessment.Wind

Each field is a direct checklist item, boolean unless stated.

- `BRIAssessment.Wind.SiteExposureAssessed`
"Site wind exposure (terrain/topography) has been explicitly assessed."
- `BRIAssessment.Wind.OrientationMitigatesWind`
"Building orientation/massing selected to reduce wind pressure where practicable."
- `BRIAssessment.Wind.StructureDesignedForHazardWind`
"Primary structure designed to required wind speed for mapped hazard class."
- `BRIAssessment.Wind.ContinuousLoadPathProvided`
"Continuous load path from roof to foundation for wind loads."
- `BRIAssessment.Wind.LateralBracingAdequate`
"Lateral bracing / shear systems appropriate for design wind loads."
- `BRIAssessment.Wind.RoofRatedForDesignWind`
"Roof system (covering, fixings, edges) rated for design wind speed."

- `BRIAssessment.Wind.RoofEdgeDetailWindResistant`
“Roof edges/parapets/eaves detailed to resist uplift.”
- `BRIAssessment.Wind.CladdingRatedForDesignWind`
“External cladding rated/anchored for design wind pressures.”
- `BRIAssessment.Wind.OpeningsWindResistant`
“External doors/windows in wind-exposed façades are rated or protected.”
- `BRIAssessment.Wind.LargeDoorsReinforced`
“Large doors (garages/loading bays) reinforced or braced for wind.”
- `BRIAssessment.Wind.RooftopEquipmentAnchored`
“Rooftop equipment anchored to resist design wind speeds.”
- `BRIAssessment.Wind.FacadeAttachmentsAnchored`
“Façade-mounted elements (signage, louvers, etc.) anchored for wind loads.”

All of these are new fields; there are no direct equivalents in the existing CDM.^[131] ^[129]

C. `BRIAssessment.Flood (Water)`

These are assessment outputs; some map conceptually to existing data but are more explicit.^[133]
^[131]

- `BRIAssessment.Flood.SiteFloodHazardAssessed`
“Site flood hazard explicitly assessed (all relevant perils).”
- `BRIAssessment.Flood.HighRiskZoneAvoidedOrJustified`
“High flood-risk locations avoided or formally justified and mitigated.”
- `BRIAssessment.Flood.FinishedFloorAboveDesignFlood`
“Finished floor of occupied spaces is above design flood level.”
- `BRIAssessment.Flood.OccupiedLevelsElevated`
“Primary occupied levels elevated (e.g. stilts/raised plinth) in high flood risk.”
- `BRIAssessment.Flood.BasementFloodStrategy (enum)`
Values like: none / no basement / flood-resistant basement / deliberately floodable with protection.
- `BRIAssessment.Flood.PermanentFloodProofingAtEntries`
“Permanent flood-resistant construction at typical entries.”
- `BRIAssessment.Flood.DeployableBarriersProvided`
“Deployable barriers/gates at all flood-exposed openings below design level.”
- `BRIAssessment.Flood.BackflowPreventionInstalled`
“Backflow prevention devices on relevant drains/sewers.”
- `BRIAssessment.Flood.OnsiteDrainageSizedForDesignStorm`
“On-site drainage sized for design storm.”
- `BRIAssessment.Flood.OverlandFlowPathsMaintained`
“Positive overland flow paths away from buildings maintained.”
- `BRIAssessment.Flood.PermeableOrRetentionMeasures`
“Permeable surfaces or retention features used for runoff.”

- `BRIAssessment.Flood.ElectricalSystemsAboveFlood`
“Main electrical/switchgear above design flood level or protected.”
- `BRIAssessment.Flood.MechanicalSystemsAboveFlood`
“Main mechanical plant above design flood level or protected.”
- `BRIAssessment.Flood.FuelAndHazardousStoresProtected`
“Fuel/hazardous stores secured against flotation/leakage in floods.”
- `BRIAssessment.Flood.SumpPumpsInstalled`
This can be **logically linked** to `ProtectionMeasures.ResilienceMeasures.SumpPump`, but keep it as a separate assessment flag so the checklist can say “yes, this BRI requirement is met” even if your raw data field exists.^[129]

You already have `FloodGates`, `FloodBarriers`, `SumpPump`, `FloodWarningSystem`; you can link:

- When `FloodGates == True`, you set `BRIAssessment.Flood.DeployableBarriersProvided = True`, etc., but the assessment field is still its own outcome.^{[130] [129]}

D. `BRIAssessment.Fire`

- `BRIAssessment.Fire.StructuralFireResistanceAdequate`
- `BRIAssessment.Fire.CompartmentsProvided`
“Compartmentation with rated walls/floors/doors provided.”
- `BRIAssessment.Fire.FireStoppingAtPenetrations`
- `BRIAssessment.Fire.ExternalMaterialsFireResistant`
- `BRIAssessment.FireAutomaticDetectionInstalled`
- `BRIAssessment.Fire.SuppressionSystemsInstalled`
- `BRIAssessment.Fire.WildfireDefensibleSpace`
- `BRIAssessment.Fire.WildfireNonCombustiblePerimeter`

All of these are new (no current equivalents).^[131]

E. `BRIAssessment.Geoseismic`

- `BRIAssessment.Geoseismic.SiteGeotechnicalAssessed`
- `BRIAssessment.Geoseismic.HighRiskGroundAvoidedOrMitigated`
- `BRIAssessment.Geoseismic.StructureMeetsSeismicCode`
- `BRIAssessment.Geoseismic.SeismicDetailingToStandard`
- `BRIAssessment.Geoseismic.StructuralRegularityAdequate`
- `BRIAssessment.Geoseismic.FoundationSuitableForHazard`
- `BRIAssessment.Geoseismic.LiquefactionMitigationProvided`
- `BRIAssessment.Geoseismic.NonstructuralComponentsAnchored`
- `BRIAssessment.Geoseismic.HeavyEquipmentAnchored`

All of these are new assessment fields; they use building information conceptually but do not duplicate it. ^[132] ^[131]

F. BRIAssessment.Continuity (for “+”)

- BRIAssessment.Continuity.BackupPowerInstalled
- BRIAssessment.Continuity.BackupPowerProtectedFromHazard
- BRIAssessment.Continuity.BackupWaterSupplyProvided
- BRIAssessment.Continuity.WaterSystemsProtectedFromHazard
- BRIAssessment.Continuity.TelecomRedundancyProvided
- BRIAssessment.Continuity.CriticalITProtected
- BRIAssessment.Continuity.AccessRouteResilient
- BRIAssessment.Continuity.AccessDesignConsidersHazards
- BRIAssessment.Continuity.BusinessContinuityPlanInPlace
- BRIAssessment.Continuity.EmergencyProceduresTested

All new; driven by operational design and management rather than core property attributes. ^[133]
^[131]

3. How this stays “assessment-only”

- The only new section is BRIAssessment, with sub-sections HazardProfile, Wind, Flood, Fire, Geoseismic, Continuity.
- It contains only:
 - Hazard classes (derived from existing data or external hazard maps).
 - Checklist booleans/enums that state whether each required/advanced measure is in place.
 - No fields from Header, Location, Valuation, etc. ^[129]

Your flow then is:

1. Populate PropertyHeader, Construction, Location, ProtectionMeasures etc. as today. ^[129]
2. Run an assessment routine that:
 - Pulls external hazard data and existing CDM fields.
 - Fills the BRIAssessment.* fields according to your BRI-style logic.
3. Use BRIAssessment.* plus a scoring rule to generate a BRI.Rating if you later choose to add BRI.Rating.Score / BRI.Rating.Category. ^[132] ^[131]

If you'd like, next step I can write a compact pseudo-schema (JSON/YAML style) for the BRIAssessment section so your engineers can drop it straight into schema-3.py.

can you create a `BRI.py` python script that I can include in `src/port/cdm/property` package, will rewire the `init.py`

I created a `BRI.py` module you can drop into `src/port/cdm/property`. It defines an assessment-only `BRIAssessment` schema, a required-fields list, helpers to extend the existing property schema, a stub builder, a lightweight derivation helper from current CDM fields, and a validator that checks only the new BRI section. [\[134\]](#) [\[135\]](#)

What's in it

The new module keeps all header/location/property facts in the existing schema and adds only assessment data under `BRIAssessment`, with sub-sections for `HazardProfile`, `Wind`, `Flood`, `Fire`, `Geoseismic`, `Continuity`, and `Rating`. It also maps the current resilience fields you already have—such as `FloodGates`, `SumpPump`, and `FloodWarningSystem`—into corresponding BRI assessment fields without duplicating property header information. [\[135\]](#) [\[136\]](#)

How to wire it

You can import `BRI_ASSESSMENT_SCHEMA` or call `extend_property_schema_with_bri (PROPERTY_SCHEMA)` to append the new section to your existing property schema. If you want a blank assessment payload for UI/forms, use `build_bri_assessment_stub()`, and if you want assessment-only validation, use `validate_bri_assessment(property_data)`. [\[137\]](#) [\[135\]](#)

Notes

I kept the implementation separate so you can rewire `__init__.py` and decide later whether to merge BRI validation into the current `validator.py` or keep it as a distinct step. The derivation helper is intentionally conservative: it only auto-populates fields that can be inferred from the current CDM, leaving the rest as explicit assessor inputs. [\[136\]](#) [\[134\]](#) [\[135\]](#) [\[137\]](#)



So all of the details now sit inside resilience. What I don't want to do is have the field BRI specific in the naming, only that they are available for a BRI index aggregator. Let's have a look at everything under resilience section and sort by function rather by utility, so for example if there is a check for use of concrete, then it sits with the building assessment rather than under a BRI validation. What we can do later is create BRI type reports, but right now let's stick to the ethos of the CDM that is utility agnostic and focuses purely on gathering information.

Yes — that's the right direction. The CDM should stay domain-descriptive, not report-descriptive: collect factual building and resilience information once, then let BRI or any other downstream framework consume it later.^[138] ^[139]

Proposed approach

Instead of a `BRIAssessment` section, I'd recommend reshaping `ProtectionMeasures.ResilienceMeasures` into functional sub-sections such as `BuildingAssessment`, `FloodProtection`, `FireProtection`, `GeoseismicProtection`, `ContinuityMeasures`, and possibly `SiteAndDrainage`. That keeps the schema utility-agnostic while still making the data rich enough for a BRI index aggregator, because the aggregator can map from these neutral facts to its own scoring logic later.^[140] ^[138]

What stays where

Fields that describe the building's intrinsic characteristics — like construction type, foundation type, number of storeys, basement presence, floor level, condition, and any future material or structural checks such as concrete use — should live with the building/construction side of the CDM, not under a framework-named validator or report section. Fields that describe installed protections or preparedness — like flood gates, barriers, sump pumps, warning systems, backup power, emergency procedures, and protected plant — belong in resilience/protection sections because they are measures applied to the asset rather than identity facts about the asset.^[140] ^[138]

Suggested resilience structure

A clean utility-agnostic layout would look like this:^[138]

- `ProtectionMeasures.BuildingAssessment` — assessed structural robustness, wind design, seismic detailing, fire resistance, material durability.
- `ProtectionMeasures.SiteAndDrainage` — site flood strategy, overland flow paths, drainage capacity, permeable surfaces, retention measures.

- `ProtectionMeasures.FloodProtection` — flood gates, barriers, backflow prevention, elevated services, sump pumps, floodproofing.
- `ProtectionMeasures.FireProtection` — detection, suppression, compartmentation, fire-stopping, defensible space.
- `ProtectionMeasures.GeoseismicProtection` — anchoring, liquefaction mitigation, slope/ground mitigation, non-structural bracing.
- `ProtectionMeasures.ContinuityMeasures` — backup power, backup water, telecom redundancy, business continuity planning, emergency testing.

That structure is closer to the current CDM ethos because it organizes by *what the information is about*, not *what report will later use it*.^[138]

Practical next step

The script I produced should therefore be revised: remove the BRI naming entirely, fold the fields into a neutral resilience schema, and only keep helper mappings or report builders as separate downstream utilities. The current CDM already hints at this pattern with `ProtectionMeasures.RiskAssessment` and `ProtectionMeasures.ResilienceMeasures`, so this would be an extension of the existing design rather than a conceptual break.^{[140] [138]}

I can now rewrite that module as a neutral `resilience.py` or `assessment.py` for `src/port/cdm/property`, with sections grouped by function and no BRI-specific field names.

✱

OK, for each field it looks like there are menus for each which I assume has been defined by BRI, we should keep this in memory when we update the port. Good, so can you map out a complete property CDM, we can later decide how it should be broken down into manageable .py

Yes — the right way to think about it is: preserve the controlled vocabularies and menu logic where they exist, but keep the CDM itself neutral and information-gathering first. I mapped out a complete Property CDM outline that combines the current schema with a functional resilience expansion, without introducing BRI-specific naming into the data model.^{[142] [143]}

What's in the outline

The outline keeps your current top-level structure of `PropertyHeader`, `ProtectionMeasures`, and `TransactionHistory`, then expands `ProtectionMeasures.ResilienceMeasures` into functional groups such as `BuildingAssessment`, `SiteAndDrainage`, `FloodProtection`, `FireProtection`, `ContinuityMeasures`, plus a neutral `HazardProfile` block. That means fields about the building itself stay with building assessment, while installed protections and continuity arrangements sit in protection/continuity sections rather than in any framework-specific namespace.^{[143] [142]}

Menu logic

I also flagged where menu-based fields are likely preferable to simple booleans, because your current schema already uses controlled options heavily and the resilience expansion should follow the same pattern. In particular, fields like hazard classes, basement flood strategy, and some compliance-style checks are better treated as enums now or later, so you can preserve the richer option logic that a BRI-style aggregator may need without polluting the CDM with BRI language.^[142]

Next step

The outline is now in a single structured reference document you can use to decide how to break it into manageable `.py` modules later. The logical follow-on would be to turn this into a concrete schema plan, for example splitting it into `schema.py`, `resilience.py`, `mapping.py`, and `validator.py` responsibilities while keeping the top-level CDM coherent.^[142]

✱

Under Risk assessment, we will need different assessment by governing body, so include a BRI Rating, that looks like AA, A, B, NR. We need to be specific of what risk ratings we are now talking about, so its been a while, I assume there is an overall resilience rating or flood-risk rating, can you check

Yes — you should separate these clearly. In your current Property CDM, `ProtectionMeasures.RiskAssessment.RiskRating` reads as a generic risk field, while the existing flood-specific risk concepts already live under `PropertyHeader.RiskAssessment` as `EAFloodZone`, `OverallFloodRisk`, and `FloodRiskType`. IFC's BRI uses a distinct overall resilience rating scheme, so that should be named separately rather than overloaded into the generic `RiskRating` field.^{[144] [145] [146]}

What BRI calls it

The official Building Resilience Index uses a building-level resilience rating, not a flood-risk rating, and public IFC materials show the current letter scale as AA, A, B, NR, with a possible + modifier for operational continuity in some materials. The brochure says the rating is a standardized letter grade used to disclose the resilience of a building, while the app/user guide explains that the overall rating is based on the “weakest link” across applicable hazard categories.^{[146] [147] [144]}

There is some version drift in public documents: the older World Bank user guide shows A+, A, B, C, R, whereas the newer brochure and webinar materials use AA, A, B, NR. If you are aligning with the more recent public BRI wording you mentioned, then AA / A / B / NR is the right controlled vocabulary to retain in memory for later use.^{[148] [147] [146]}

What your current risk fields mean

In your current schema, these are the relevant existing fields:^[145]

- `PropertyHeader.RiskAssessment.EAFloodZone` = flood hazard zoning.
- `PropertyHeader.RiskAssessment.OverallFloodRisk` = overall flood risk level.
- `PropertyHeader.RiskAssessment.FloodRiskType` = type of flood risk.
- `ProtectionMeasures.RiskAssessment.RiskRating` = generic risk rating, currently not explicitly tied to BRI.

So yes, there is already an overall flood risk concept in the CDM, but there is **not yet an explicitly named overall resilience rating field** distinct from flood.^[145]

Recommended naming

I'd recommend keeping `ProtectionMeasures.RiskAssessment.RiskRating` only if you intend it to mean a broad internal or insurer-style risk rating. Then add a separate governing-body specific ratings subsection, for example:^[145]

- `ProtectionMeasures.RiskAssessment.GoverningBodyRatings.BRIRating`
- `ProtectionMeasures.RiskAssessment.GoverningBodyRatings.BRIRatingVersion`
- `ProtectionMeasures.RiskAssessment.GoverningBodyRatings.BRIRatingDate`

with `BRIRating` options set to AA, A, B, NR for the newer IFC scale.^[147] ^[148]

That gives you clean separation between:

- **Flood risk:** hazard/risk to the property from flooding.^[145]
- **Generic risk rating:** your own or insurer/internal risk label.^[145]
- **BRI rating:** an external resilience classification from a governing framework.^[146] ^[147]

Practical schema interpretation

If you want the CDM to stay utility-agnostic, the safest pattern is:

- Keep hazard-specific risk fields under `PropertyHeader.RiskAssessment`.^[145]
- Keep neutral internal/composite risk fields under `ProtectionMeasures.RiskAssessment`.^[145]
- Add a sub-section for external framework ratings, such as `GoverningBodyRatings`, where BRI is one entry today and others can be added later.^[147] ^[146]

A concrete example would be:

```
ProtectionMeasures
  RiskAssessment
    InsurancePremium
    InternalRiskRating
    GoverningBodyRatings
      BRIRating
      BRIRatingVersion
      BRIRatingDate
```

That structure avoids confusing a flood-risk label with a resilience label and leaves room for future ratings by other bodies. [\[146\]](#) [\[145\]](#)

Would you like me to turn that into a revised schema outline showing exactly where BRIRating would sit alongside OverallFloodRisk and any renamed generic risk fields?

✱

I suggest we use RiskRating for insurance, so be more specific that its used by the local insurance body. Bit of a placeholder, but makes sense to differentiate from the BRI

Yes, that makes sense. Your existing RiskRating can be repurposed as an insurance-facing field, while BRIRating remains a distinct external resilience classification, because IFC positions BRI as a resilience framework that can influence insurance outcomes rather than replace insurer risk judgements. [\[162\]](#) [\[163\]](#) [\[164\]](#)

Recommended distinction

I'd make the distinction explicit in the schema naming :

- ProtectionMeasures.RiskAssessment.InsuranceRiskRating
- ProtectionMeasures.RiskAssessment.InsuranceRiskRatingBody
- ProtectionMeasures.RiskAssessment.InsurancePremium
- ProtectionMeasures.RiskAssessment.GoverningBodyRatings.BRIRating

This is cleaner than leaving RiskRating ambiguous, because BRI is an overall resilience rating and your insurer field is really a local or market-specific underwriting/risk-quality label. [\[163\]](#) [\[165\]](#)

Why this works

Your current schema already pairs InsurancePremium with RiskRating, which strongly suggests that field belongs with insurance logic rather than resilience scoring. That also fits IFC's messaging: BRI helps owners disclose resilience and potentially negotiate better premiums or broader coverage, which means BRI and insurance ratings are related but not the same thing. [\[166\]](#) [\[165\]](#) [\[162\]](#) [\[163\]](#)

Suggested placeholder wording

If you want a practical placeholder that stays utility-agnostic enough for now, I'd use:

- InsuranceRiskRating — "Risk rating assigned by the relevant local insurance market, insurer, or underwriting body."
- InsuranceRiskRatingBody — "Name of insurer, broker, pool, or local insurance authority that assigned the insurance risk rating."

- BRIRating — “Building resilience rating assigned under the IFC Building Resilience Index methodology.”

That gives you one field for local insurance judgement and one for a standardized resilience framework, with no collision in meaning.^[164] ^[163]

Schema direction

So yes: rename the old generic RiskRating concept to something insurance-specific, then add BRIRating separately under a governing-body or external-ratings subsection. That will make the model much easier to understand later when you introduce other frameworks besides BRI.^[165]

A sensible target structure would be:

```
ProtectionMeasures
  RiskAssessment
    InsurancePremium
    InsuranceRiskRating
    InsuranceRiskRatingBody
  GoverningBodyRatings
    BRIRating
    BRIRatingDate
    BRIRatingVersion
```

Would you like me to produce a revised full CDM outline with that naming change applied throughout?

✱

I would use the same structure for BRI, so InsuranceBodyRatings, then sub, InsuranceRating, InsuranceDate, InsuranceRatingVersion. I would leave out Insurance Premium as that is financial information and should go elsewhere.

Yes, that structure is cleaner. Since BRI is a separate rating framework and insurance is a separate local/market judgement, mirroring the same sub-structure for both makes the schema more consistent and easier to extend.^[178] ^[179]

Suggested structure

I'd recommend moving financial data out of this section and defining ratings like this:

```
ProtectionMeasures
  RiskAssessment
    InsuranceBodyRatings
      InsuranceRating
      InsuranceDate
      InsuranceRatingVersion
```

```
InsuranceRatingBody
GoverningBodyRatings
BRIRating
BRIDate
BRIRatingVersion
BRIRatingBody
```

That works because IFC presents BRI as a standardized resilience rating used for disclosure, while insurance entities use their own underwriting or risk-quality views that may reference BRI but are not the same thing.^[180] ^[179]

Why remove premium

Leaving out InsurancePremium from RiskAssessment is sensible if you want this section to stay focused on assessment outputs rather than commercial terms. Premium is financial information, so it would sit better in a pricing, coverage, or insurance-financial section later, whereas ratings and rating metadata are assessment/disclosure data.^[181]

Naming refinement

To keep the model self-explanatory, I'd suggest slightly fuller names inside those subsections:

- InsuranceBodyRatings.InsuranceRating
- InsuranceBodyRatings.InsuranceDate
- InsuranceBodyRatings.InsuranceRatingVersion
- InsuranceBodyRatings.InsuranceRatingBody

and

- GoverningBodyRatings.BRIRating
- GoverningBodyRatings.BRIDate
- GoverningBodyRatings.BRIRatingVersion
- GoverningBodyRatings.BRIRatingBody

The BRI rating scale should use the current public letter grades AA, A, B, NR, with optional future support for a + continuity modifier if you decide to capture that later.^[179]

CDM implication

So the main change from the current schema is: RiskAssessment becomes the home for assessment classifications and metadata, not premiums, while hazard facts like flood zone and flood risk remain under PropertyHeader.RiskAssessment. That gives you a clear split between hazard/exposure facts, assessment outcomes, and financial/commercial data.^[181]

I can now produce a revised full CDM outline reflecting this exact structure and naming.



I would replace BRIRatingBody with BRIRatingAgent as BRI is the rating body.

Yes — BRIRatingAgent is the better name. BRI is the framework or rating body, while the organization or party applying, verifying, or issuing the rating is a separate actor, and public IFC/Bureau Veritas materials explicitly refer to third-party verifier partners such as Bureau Veritas and RINA.^[194] ^[195] ^[196]

Recommended naming

I'd use this structure:

```
ProtectionMeasures
  RiskAssessment
    InsuranceBodyRatings
      InsuranceRating
      InsuranceDate
      InsuranceRatingVersion
      InsuranceRatingBody
    GoverningBodyRatings
      BRIRating
      BRIDate
      BRIRatingVersion
      BRIRatingAgent
```

That keeps the pattern symmetrical while correctly distinguishing:

- BRI as the framework behind the rating, and
- BRIRatingAgent as the verifier / certifier / assessment agent involved in that rating.^[195] ^[196]

Why “agent” fits

IFC's recent announcements say Bureau Veritas serves as a BRI verification partner or global verifier, rather than replacing BRI itself as the rating framework. So BRIRatingBody would blur the distinction, while BRIRatingAgent cleanly captures the entity that performed, verified, or administered the rating.^[196] ^[197] ^[194] ^[195]

Slight refinement

If you want the naming to stay flexible across schemes, you could eventually standardize on:

- ...Rating
- ...Date
- ...RatingVersion
- ...RatingAgent

for both insurance and governing-body ratings. That would mean InsuranceRatingAgent instead of InsuranceRatingBody, but your current suggestion is already structurally sound.^[195] ^[196]

So yes — I agree with changing BRIRatingBody to BRIRatingAgent.^[197] ^[194]

I can now produce the revised full CDM outline with:

- InsuranceBodyRatings
- GoverningBodyRatings
- BRIRatingAgent
- no InsurancePremium in RiskAssessment

✱

Can you produce a revised full CDM for review

I produced a revised full Property CDM review document reflecting the latest decisions: resilience grouped by function, InsurancePremium removed from RiskAssessment, parallel insurance and governing-body rating structures, and BRIRatingAgent replacing BRIRatingBody. [\[208\]](#) [\[209\]](#)

What changed

The revised outline keeps PropertyHeader.RiskAssessment for hazard/exposure facts such as flood zone and overall flood risk, while ProtectionMeasures.RiskAssessment now holds only assessment and rating metadata. It also introduces InsuranceBodyRatings and GoverningBodyRatings as parallel sub-structures, which should make future extension much easier. [\[209\]](#) [\[210\]](#) [\[208\]](#)

Main review points

The key review items are whether HazardProfile should remain under ProtectionMeasures or move closer to PropertyHeader.RiskAssessment, whether rating subsections should later support multiple named schemes, and whether some resilience checks should be split further by hazard type. I also flagged which new fields are good menu candidates versus fields that can start as booleans and later evolve into richer enums. [\[208\]](#)

✱

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